

Sequential Anytime-Valid Inference (SAVI) Using E-Processes (Part II)

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Overview

1. Recap of part I

- 1.1 Motivation, setup, definitions
- 1.2 Validity: e-processes under $\mathcal{P}$
- 1.3 Efficiency: e-processes under $\mathcal{Q}$

2. Constructing e-processes (valid for composite $\mathcal{P}$)

- 2.1 Universal inference (UI)
- 2.2 Betting via sequential e-variables
- 2.3 Mixing fixed-sample-size e-variables

3. Further discussions

- 3.1 Caveat: pathologies of e-power
- 3.2 E-processes avoid reasoning about hypothetical worlds

4. Summary

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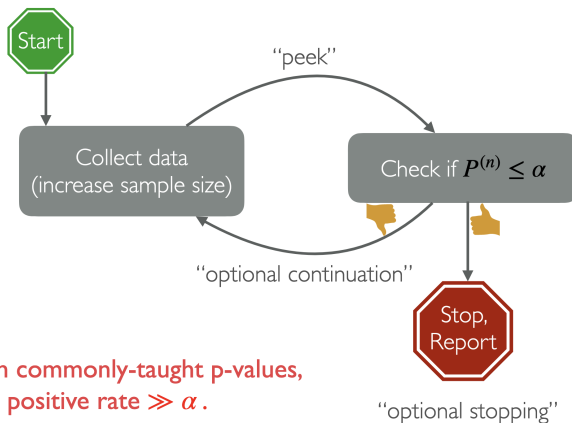
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Motivation for SAVI (recap)

What is the problem with continuous monitoring?



With commonly-taught p-values,
false positive rate $\gg \alpha$.

Setup (recap)

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Natural filtration: for $t \in \{0, 1, \dots\}$, $\mathcal{F}_t = \sigma(X_1, \dots, X_t)$, $\mathcal{F}_0 = \{\emptyset, \Omega\}$, $\mathcal{F} := (\mathcal{F}_t)_{t \geq 0}$

Process: sequence of r.v.s $Y = (Y_t)_{t \geq 0}$ adapted to $\mathcal{F}$, i.e., every Y_t is $\mathcal{F}_t$ -measurable.

Y_t is predictable if Y_t is $\mathcal{F}_{t-1}$ -measurable ($t \geq 1$)

Stopping time (or rule) τ : nonnegative integer-valued r.v. s.t. $\{\tau \leq t\} \in \mathcal{F}_t$ for each $t \geq 0$.

Level- α sequential test for $\mathcal{P}$: a binary process $\phi = (\phi_t)_{t \geq 1}$ with

$$\sup_{\mathbb{P} \in \mathcal{P}} \mathbb{P}(\exists t \geq 1 : \phi_t = 1) \leq \alpha$$

Test supermartingales & e-processes (recap)

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Test supermartingale for $\mathcal{P}$. For every $\mathbb{P} \in \mathcal{P}$, $M = (M_t)_{t \geq 0}$ satisfies

- $M_t \geq 0$ $\mathbb{P}$ -a.s.,
- $\mathbb{E}^{\mathbb{P}}[M_t \mid \mathcal{F}_{t-1}] \leq M_{t-1}$, and
- $\mathbb{E}^{\mathbb{P}}[M_0] \leq 1$.

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- $\mathbb{E}^{\mathbb{P}}[M_0] \leq 1$.

E-process for $\mathcal{P}$. A sequence of e-values $E = (E_t)_{t \geq 0}$ adapted to $\mathcal{F}$ and either (equivalently):

- $\mathbb{E}^{\mathbb{P}}[E_\tau] \leq 1$ for **every** stopping time τ and $\mathbb{P} \in \mathcal{P}$; or
- $\exists$ a test supermartingale family $(M^{\mathbb{P}})_{\mathbb{P} \in \mathcal{P}}$ with $E_t \leq M_t^{\mathbb{P}}$, $\mathbb{P}$ -a.s. for every $t \geq 0$ and $\mathbb{P} \in \mathcal{P}$.

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Optional stopping & Ville's inequality (recap)

Optional stopping. If M is a test supermartingale for $\mathbb{P}$, then for any stopping time τ and $\sigma \leq \tau$:

$$\mathbb{E}^{\mathbb{P}}[M_{\tau} \mid \mathcal{F}_{\sigma}] \leq M_{\sigma} \implies \mathbb{E}^{\mathbb{P}}[M_{\tau}] \leq \mathbb{E}^{\mathbb{P}}[M_0] \leq 1.$$

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Ville's inequality. If M is a non-negative supermartingale for $\mathbb{P}$, then for any $x > 0$,

$$\mathbb{P}(\exists t \in \mathbb{N} : M_t \geq x) \leq \frac{\mathbb{E}^{\mathbb{P}}[M_0]}{x}$$

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Ville's inequality. If M is an e-process for $\mathcal{P}$, then for any $\alpha \in (0, 1]$,

$$\mathbb{P} \left(\exists t \in \mathbb{N} : M_t \geq \frac{1}{\alpha} \right) \leq \alpha$$

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Proposition

Every **level- α sequential test** for $\mathcal{P}$ is a **e-process** for $\mathcal{P}$ thresholded at $1/\alpha$.

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Optional continuation. If M^A, M^B are e-processes for $\mathcal{P}$ and $M_t^B = 1$ on $\{t \leq \tau\}$, then $M = (M_t)$ with $M_t := M_{\tau \wedge t}^A M_t^B$ is an e-process for $\mathcal{P}$.

Nontrivial e-processes exist

* : no reference measure

Any $\mathcal{P}$ that is sequentially testable

Exchangeability* (in original filtration)

T-test (in original filtration)

Nontrivial test supermartingales exist

SubGaussian distributions* (or any bounded MGF)

Robust, heavy-tailed mean estimation*

Nontrivial test martingales exist

Testing symmetry*

Two-sample testing*

Bounded means*

T-test (in shrunk filtration)

Exchangeability* (in shrunk filtration)

Independence testing* (in shrunk filtration)

Figure: Hierarchy of tools for constructing sequential tests: nontrivial e-processes (outermost), test supermartingales (middle), and test martingales (innermost), with examples of what each can test.

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Log-optimality (testing $\mathbb{P}$ vs $\mathbb{Q}$) (recap)

The LR process M^* given by $M_0^* = 1$ and $M_t^* = \frac{d\mathbb{Q}|_{\mathcal{F}_t}}{d\mathbb{P}|_{\mathcal{F}_t}}(X_1, \dots, X_t)$ is a test martingale for $\mathbb{P}$.

Theorem (Log-optimality)

For any stopping time τ that is finite $\mathbb{Q}$ -a.s. and any e-process M for $\mathbb{P}$:

$$\mathbb{E}^{\mathbb{Q}}[\log M_{\tau}^*] \geq \mathbb{E}^{\mathbb{Q}}[\log M_{\tau}].$$

$\implies$ LR processes are useful benchmark for testing against composite alternatives.

Asymptotic log-optimality (testing $\mathbb{P}$ vs $\mathcal{Q}$) (recap)

Definition

An e-process M is **asymptotically log-optimal** for $\mathbb{P}$ against $\mathcal{Q}$ if for **every** $\mathbb{Q} \in \mathcal{Q}$,

$$\lim_{t \rightarrow \infty} \frac{1}{t} \left(\log M_t - \log M_t^{\mathbb{Q}} \right) \geq 0 \quad \text{in } L^1\text{-convergence under } \mathbb{Q}$$

where $M^{\mathbb{Q}}$ is the LR process of $\mathbb{Q}$ to $\mathbb{P}$.

$\implies$ covers any e-process M that **grows an $e^{o(t)}$ factor slower** than $M^{\mathbb{Q}}$.

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Testing $\mathbb{P}$ against $\mathcal{Q} = \{\mathbb{Q}_{\theta} : \theta \in \Theta_1\}$ with iid data: $M_0 = 1$, and with predictable $\hat{\theta}_{i-1}$, use

- **Plug-in LR:** $M_t = \prod_{i=1}^t \frac{q_{\hat{\theta}_{i-1}}(X_i)}{p(X_i)}$, or
- **Mixture LR:** $M_t = \int_{\Theta_1} \prod_{i=1}^t \frac{q_{\theta}(X_i)}{p(X_i)} \nu(d\theta)$, ν a prior on Θ_1 .

Plug-in is asymptotically log-optimal when $\theta_i \rightarrow \theta$ under $\mathbb{Q}_{\theta}$ in a suitable sense, given log-LR is concave, score function has bounded variance.

Example: Given iid data from $N(\theta^\dagger, 1)$, goal is to test $H_0 : \theta^\dagger = 0$ vs $H_1 : \theta^\dagger > 0$. For illustration, take $\theta^\dagger = 0.3$.

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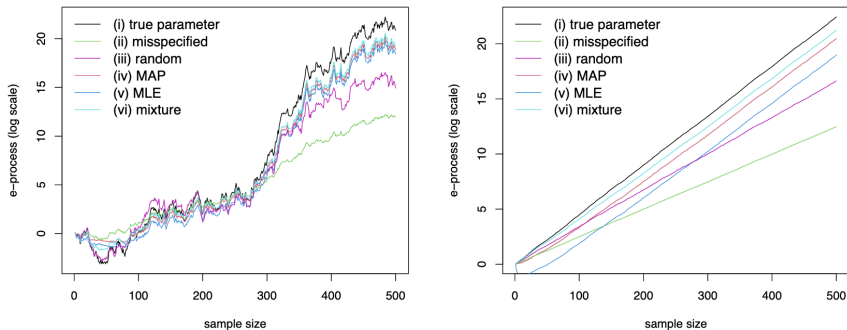


Figure: few ways of constructing e-values from LR processes. Left: one run. Right: average of 1000 runs.

- (i) true parameter: choose $\theta_i = \theta^\dagger = 0.3$
- (ii) misspecified: choose $\theta_i = 0.1$
- (iii) random: take iid θ_i from $U[0, 0.5]$
- (iv) MAP: choose θ_i by the MAP estimator with prior $\theta \sim N(0.1, 0.2^2)$
- (v) MLE: choose θ_i with $\theta_1 := 0.1$ and θ_i the MLE of θ based on $X_1, \dots, X_{i-1}$
- (vi) mixture: compute a mixture of M each with a fixed $\theta_i \in [0, 0.6]$, uniformly weighted via a discrete grid

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Universal Inference (UI) e-variable & e-process

Goal: build an e-variable (or an e-process) to test a composite null $\mathcal{P}$ against composite alternative $\mathcal{Q}$ **under no regularity conditions**.

Recipe: a reduction from hypothesis testing to MLE under $\mathcal{P}$.

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Theorem (UI/Split LR e-variable)

Split data $X_{1:n}$ into $D_0 \perp D_1$, define

$$E = \prod_{i \in D_0} \frac{\hat{q}_1(X_i)}{\hat{p}_0(X_i)}, \quad \hat{q}_1 \in \mathcal{Q} \text{ learned from } D_1, \quad \hat{p}_0 \in \mathcal{P} \text{ is the MLE on } D_0,$$

then E is an e-variable for $\mathcal{P}$.

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Proposition (UI e-process)

Define $E_0 = 1$ and

$$E_t = \prod_{i=1}^t \frac{\hat{q}_{i-1}(X_i)}{\hat{p}_t(X_i)} \text{ for } t \geq 1, \quad \hat{q}_{i-1} \in \mathcal{Q} \text{ learned from } X_{1:i-1}, \quad \hat{p}_t \in \mathcal{P} \text{ is the MLE on } X_{1:t},$$

then $E = (E_t)$ is an e-process for $\mathcal{P}$.

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then $E = (E_t)$ is an e-process for $\mathcal{P}$.

Proof. For every $\mathbb{P} \in \mathcal{P}$, let $M_0^{\mathbb{P}} = E_0 = 1$, and

$$M_t^{\mathbb{P}} := \prod_{i=1}^t \frac{\hat{q}_{i-1}(X_i)}{p(X_i)} \geq \prod_{i=1}^t \frac{\hat{q}_{i-1}(X_i)}{\hat{p}_t(X_i)} = E_t \text{ for } t \geq 1.$$

Then $M_t^{\mathbb{P}}$ is a test martingale for $\mathbb{P}$.

$\Rightarrow$ optional stopping: $\mathbb{E}^{\mathbb{P}}[M_{\tau}^{\mathbb{P}}] \leq \mathbb{E}^{\mathbb{P}}[M_0^{\mathbb{P}}] = 1 \Rightarrow \mathbb{E}^{\mathbb{P}}[E_{\tau}] \leq \mathbb{E}^{\mathbb{P}}[M_{\tau}^{\mathbb{P}}] \leq 1$ for every stopping time τ .

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then $E = (E_t)$ is an e-process for $\mathcal{P}$.

Remarks.

- Learn alternative out-of-sample; fit null in-sample.
- The derived sequential test $\mathbb{1}\{E_t \geq \frac{1}{\alpha}\}$ is computationally expensive: while numerator can be updated in an online fashion, all terms in denominator need to be recalculated after observing each new data point costing $O(t)$ at step t .
- Can use mixture instead of plug-in: $E_t = \int_{\mathcal{Q}} \prod_{i=1}^t \frac{q(X_i)}{\hat{p}_t(X_i)} \nu(dq)$ for any distribution ν over $\mathcal{Q}$.

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Testing by betting

Initialize wealth $M_0 = 1$.

For $t = 1, 2, \dots$:

- Declare a bet $E_t : \mathcal{X} \rightarrow [0, \infty)$ with $\mathbb{E}^{\mathbb{P}}[\textcolor{red}{E}_t(X_t) \mid \mathcal{F}_{t-1}] \leq 1 \quad \forall \mathbb{P} \in \mathcal{P}$.
- Observe data X_t .
- Update wealth: $M_t = M_{t-1} \cdot \textcolor{red}{E}_t(X_t) = \prod_{s=1}^t E_s(X_s)$.

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Definition (Sequential e-variables)

The e-variables E_t with $t \geq 1$ for $\mathcal{P}$, adapted to the filtration $\mathcal{F}$, are *sequential* if

$$\mathbb{E}^{\mathbb{P}}[E_t \mid \underbrace{E_1, \dots, E_{t-1}}_{\mathcal{F}_{t-1}}] \leq 1, \quad \forall \mathbb{P} \in \mathcal{P}, \quad \forall t \geq 1.$$

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Proposition

If $(E_t)_{t \geq 1}$ are sequential e-variables for $\mathcal{P}$,

$$M_t = \prod_{s=1}^t E_s \quad \text{for } t \geq 1, \quad M_0 = 1,$$

is a test supermartingale (hence e-process) for $\mathcal{P}$.

Proof. $\mathbb{E}^{\mathbb{P}}[M_t \mid \mathcal{F}_{t-1}] = M_{t-1} \mathbb{E}^{\mathbb{P}}[E_t \mid \mathcal{F}_{t-1}] \leq M_{t-1}$ for every $\mathbb{P} \in \mathcal{P}$.

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Question: What are the optimal bets?

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- For simple $\mathcal{P} = \{\mathbb{P}\}$ and $\mathcal{Q} = \{\mathbb{Q}\}$, $E_t(X_t) = \frac{q(X_t|\mathcal{F}_{t-1})}{p(X_t|\mathcal{F}_{t-1})}$ ensures (M_t) is log-optimal.

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- For composite $\mathcal{P}$,
 - (i) No known analogue of the LR increments that makes (M_t) log-optimal;
 - (ii) While **numeraires** are log-optimal, they are designed for fixed sample size,
 $\implies E_t = E^{(t)*} / E^{(t-1)*}$ need not satisfy $\mathbb{E}^{\mathbb{P}}[E_t \mid \mathcal{F}_{t-1}] \leq 1$ and $(E^{(t)*})_{t \geq 1}$ is valid for the “least-favourable” $\mathbb{P}^* \in \mathcal{P}^{\circ\circ}$ via RIPr but not for all $\mathbb{P} \in \mathcal{P}$.

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 $\implies E_t = E^{(t)*} / E^{(t-1)*}$ need not satisfy $\mathbb{E}^{\mathbb{P}}[E_t \mid \mathcal{F}_{t-1}] \leq 1$ and $(E^{(t)*})_{t \geq 1}$ is valid for the “least-favourable” $\mathbb{P}^* \in \mathcal{P}^{\circ\circ}$ via RIPr but not for all $\mathbb{P} \in \mathcal{P}$.
 - (iii) **Possible solution: Avoid all-in; pick $\lambda_t \in [0, 1]$ to hedge misspecification.**

Testing by betting (generalised)

Initialize wealth $M_0 = 1$.

For $t = 1, 2, \dots$:

- Declare a bet $E_t : \mathcal{X} \rightarrow [0, \infty)$ with $\mathbb{E}^{\mathbb{P}}[E_t(X_t) \mid \mathcal{F}_{t-1}] \leq 1 \quad \forall \mathbb{P} \in \mathcal{P}$.
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- Observe data X_t .

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Proposition

$(M_t)_{t \geq 0}$ is a test supermartingale (hence e-process) for $\mathcal{P}$.

Proof. $\mathbb{E}^{\mathbb{P}}[(1 - \lambda_t) + \lambda_t E_t \mid \mathcal{F}_{t-1}] \leq (1 - \lambda_t) + \lambda_t \cdot 1 = 1$ for every $\mathbb{P} \in \mathcal{P}$.

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Definition

$(M_t)_{t \geq 0}$ is called an e-process built on $(E_t)_{t \geq 1}$.

Optimising predictable stakes $(\lambda_t)_{t \geq 1}$

Definitions

(i) For an alternative measure $\mathbb{Q}$, the **$\mathbb{Q}$ -log-optimal** e-process built on (E_t) is (M_t) with

$$\lambda_t \in \arg \max_{\lambda \in [0,1]} \mathbb{E}^{\mathbb{Q}} [\log ((1 - \lambda) + \lambda E_t) \mid \mathcal{F}_{t-1}].$$

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- (M_t) generated from method (i) is **log-optimal among e-processes built on (E_t)**

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- (M_t) generated from method (i) is **log-optimal among e-processes built on (E_t)**
- (M_t) generated from method (ii), **GRAPA**, has good e-power if (E_t) are roughly iid under $\mathbb{Q}$.

=Growth Rate Adaptive to the Particular Alternative

Empirically adaptive e-processes

Theorem

Let $(E_t)_{t \geq 1}$ be iid under the alternative distribution $\mathbb{Q}$ such that $\mathbb{E}^{\mathbb{Q}}[\log E_1]$ is finite. The empirically adaptive e-process $(M_t)_{t \geq 0}$ with $\gamma = 1$ satisfies the following:

(i) Asymptotic log-optimality in the sense that

$$\lim_{t \rightarrow \infty} \frac{1}{t} \left(\log M_t - \log M_t^{\mathbb{Q}} \right) \geq 0 \quad \text{in } L^1\text{-convergence under } \mathbb{Q}$$

for the $\mathbb{Q}$ -log-optimal e-process $(M_t^{\mathbb{Q}})_{t \geq 0}$ built on $(E_t)_{t \geq 1}$.

(ii) Consistency, i.e., if $\mathbb{E}^{\mathbb{Q}}[E_1] > 1$, then $M_t \rightarrow \infty$ $\mathbb{Q}$ -a.s. as $t \rightarrow \infty$.

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Proof.

(i) follows from LLN.

(ii) applies Theorem 3.14: for $E \geq 0$, $\mathbb{E}^{\mathbb{Q}}[E] > 1 \Leftrightarrow \exists \lambda \in [0, 1]$ s.t. $\mathbb{E}^{\mathbb{Q}}[\log((1 - \lambda) + \lambda E)] > 0$.

Empirically adaptive e-processes

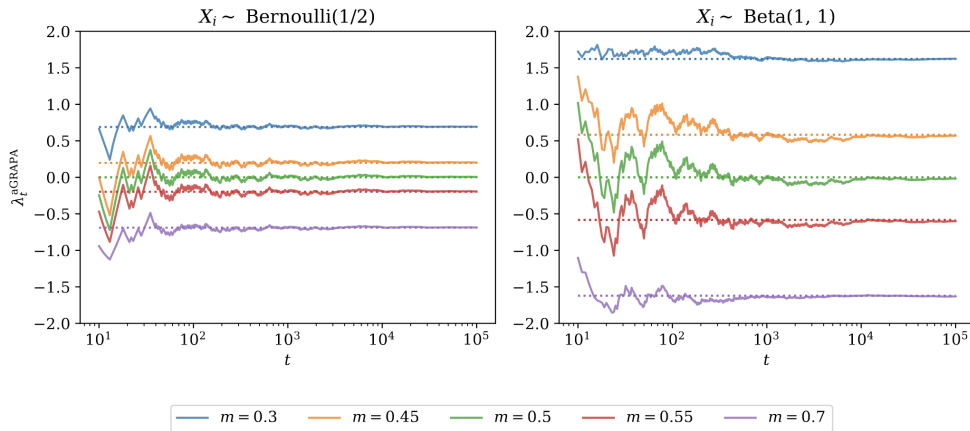


Figure: λ_t chosen using approximated GRAPA under 2 data distributions; dotted lines show oracle stakes.

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2. Constructing e-processes (valid for composite $\mathcal{P}$)

- 2.1 Universal inference (UI)
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3. Further discussions

- 3.1 Caveat: pathologies of e-power
- 3.2 E-processes avoid reasoning about hypothetical worlds

4. Summary

Mixing fixed-sample-size e-variables into e-process

For all $t \geq 1$, let $E^{(t)}$ be e-variable for $\mathcal{P}$ based on $X_{1:t}$. With any pmf w on $\mathbb{N}$, define

$$M_t = \sum_{j=1}^t w(j) E^{(j)}, \quad M_0 = 1.$$

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Proof. For any $\mathbb{P} \in \mathcal{P}$ and any random time τ ,

$$\mathbb{E}^{\mathbb{P}}[M_\tau] = \sum_{j \geq 1} w(j) \mathbb{E}^{\mathbb{P}}[E^{(j)} \cdot \mathbb{1}\{\tau \geq j\}] \leq \sum_{j \geq 1} w(j) = 1.$$

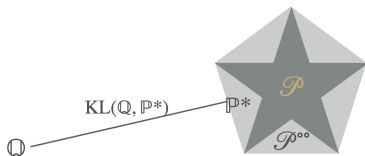
Remark. $(M_t)_{t \geq 1}$ is an increasing process, which is unusual for e-processes.

Mixing fixed-sample-size **numeraires** into e-process

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Recall:

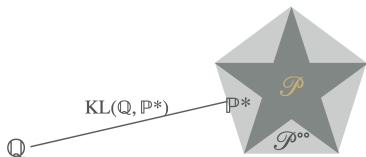
- A numeraire is a $\mathbb{Q}$ -a.s. strictly positive e-variable E^* s.t. $\mathbb{E}^{\mathbb{Q}}[E/E^*] \leq 1$ for every e-variable E . $\implies \mathbb{E}^{\mathbb{Q}}[\log(E/E^*)] \leq 0$.
- “**Composite LR**”: E^* is LR between $\mathbb{Q}$ and some element $\mathbb{P}^* \in \mathcal{P}^{\circ\circ}$ (i.e., RIPr of $\mathbb{Q}$ onto $\mathcal{P}$).



Mixing fixed-sample-size **numeraires** into e-process

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Can mix fixed-sample-size numeraires to obtain an e-process:

Popular choice of mixing pmf: $w(t) = \frac{c}{t(\log t)^2}$ with $\sum_{t \in \mathbb{N}} w(t) = 1$ and constant $c > 0$.

$$\implies \log M_t \geq \log(w(t)E^{(t)*}) = \log E^{(t)*} - \log t - 2 \log \log t + \log c$$

$$\implies \lim_{t \rightarrow \infty} \frac{1}{t} (\log M_t - \log E^{(t)*}) = 0.$$

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Caveat: pathologies of e-power

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Heavy tailed $\mathbb{Q}$ can make finite-sample e-power misleading:

Example: let $Y_1, Y_2, \dots$ be iid Pareto(1) under $\mathbb{Q}$ (i.e., $\mathbb{Q}(Y > x) = 1/x$ for $x \geq 1$), define $E_t = \exp(t^2 - Y_t)$ and set $M_t = \prod_{s=1}^t E_s$ for $t \in \mathbb{N}$. Then

$$\implies \mathbb{E}^{\mathbb{Q}}[\log E_t] = -\infty, \text{ but } \log E_t \text{ is } \mathbb{Q}\text{-a.s. finite}$$

$$\implies \mathbb{E}^{\mathbb{Q}}[\log M_t] = -\infty \text{ for every } t, \text{ while } M_t \rightarrow \infty \text{ in probability under } \mathbb{Q} \text{ (so } \mathbb{Q}(M_t \geq 1/\alpha) \rightarrow 1)$$

Takeaway: Extra caution needed when infinity is involved in the calculation of e-power. Use consistency diagnostics alongside e-power.

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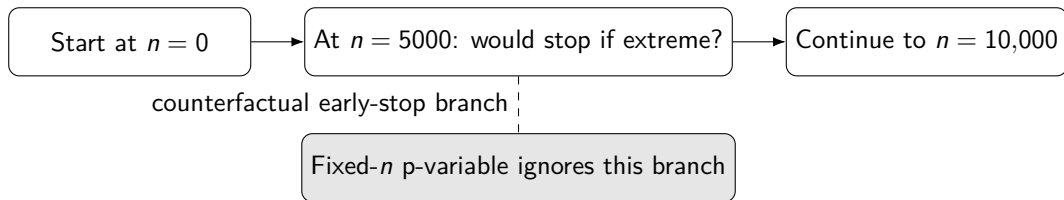
E-processes avoid reasoning about hypothetical worlds

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Even if an analyst **actually** stops at $n = 10,000$, the mere **willingness** to stop earlier if results had been extreme makes the stopping time data-dependent.

Consequence. Reasoning about the validity of p-variables requires reasoning about all possible “hypothetical worlds”.

E-process fix. Report E_τ (or threshold at $1/\alpha$); valid for **any** stopping time τ , including adaptive or hypothetical ones.



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Simple null:

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Simple null:

- LR process is log-optimal
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Composite null:

- **UI e-process:** learn alt out-of-sample, fit null in-sample; applicable to irregular problems.
- **Betting via sequential e-variables** (E_t): Tune stakes $\lambda_t \in [0, 1]$ to form empirically adaptive e-process, matching asymptotic growth rate of its oracle counterpart if (E_t) are iid.
- **Mixing fixed-sample-size numeraires:** resulting e-process matches the numeraires' asymptotic growth rate.